① for each of the following case, determine the specified properties using tables.

ⓐ for water at $P = 100\ lbf/in^2$, and $u = 500\ BTU/lb$, determine T in °F and v in ft^3/lb.

water at phase liquid vapor

when pressure is $100\ lbf/in^2$, Temperature is,

$$P_2 = 103 \quad P_1 = 89.6 \quad P_x = 100 \quad \left\{ \quad T_2 = 330 \quad T_1 = 320 \quad T_x = ? \right.$$

$$\left(\frac{P_2 - P_1}{P_2 - P_x}\right) = \left(\frac{T_2 - T_1}{T_2 - T_x}\right)$$ finding T_x

$$T_2 - T_x = \frac{(T_2 - T_1)(P_2 - P_x)}{P_2 - P_1}$$

$$-T_x = \frac{(T_2 - T_1)(P_2 - P_x)}{(P_2 - P_1)} - T_2$$

$$-T_x = \frac{(330 - 320)(103 - 100)}{(103 - 89.6)} - 330$$

$$-T_x = -327.76$$

$$\boxed{T_x = 327.76\,°F}$$

$P_2 = 103 \quad v_2 = 0.01776$
$P_1 = 89.6 \quad v_1 = 0.01775$ (liquid)
$P_x = 100 \quad v_x = ?$

$$\frac{P_2 - P_1}{P_2 - P_x} = \frac{v_2 - v_1}{v_2 - v_x}$$

$$-v_x = \frac{(v_2 - v_1)(P_2 - P_x)}{P_2 - P_1} - v_2$$

$$= \frac{(0.01776 - 0.01775)(103 - 100)}{(103 - 89.6)} - 0.01776$$

$$-v_x = -0.017757$$

$$\boxed{v_x = 0.017757\ ft^3/lb}$$

for gas v_{gas} =

$P_2 = 103 \quad v_2 = 4.312$
$P_1 = 89.60 \quad v_1 = 4.919$
$P_x = 100 \quad v_x = ?$

$$\frac{P_2 - P_1}{P_2 - P_x} = \frac{v_2 - v_1}{v_2 - v_x}$$

$$\frac{(v_2 - v_1)(P_2 - P_x)}{(P_2 - P_1)} = v_2 - v_x = \frac{(4.312 - 4.919)(103 - 100)}{(103 - 89.6)}$$

$$v_2 - v_x = -0.1359$$

$$4.312 + 0.1359 = \boxed{v_x = 4.448\ ft^3/lb}$$

Specific vol of water at liquid vapor stape for gas

Now $u = 500\ BTU/lb$ — finding specific volume

for liquid phase

for gas phase

$v_g = 0.5805\ ft^3/$

$u_2 = 508.5$, $u_1 = 485.1$, $u_x = 500$ $\left(BTU/lb\right)$ $\quad v_2 = 0.02091$, $v_1 = 0.02043$, $v_f =$ find $\left(ft^3/lb\right)$

$$\frac{u_2-u_1}{u_2-u_x} = \frac{v_2-v_1}{v_2-v_f}$$

$$v_2-v_f = \frac{(v_2-v_1)(u_2-u_x)}{(u_2-u_1)}$$

$$0.02091 - v_f = \frac{(0.0209-0.02043)(508.5-500)}{(508.5-485.1)}$$

$$0.02091 - 1.71\times10^{-4} = \boxed{v_f = 0.02074 \ ft^3/lb}$$

$v_2 = \ldots$, $v_1 = 0.6761$, $v_f =$ find $\left(ft^3/lb\right)$

$$v_2 - v_f = \frac{(v_2-v_1)(u_2-u_x)}{(u_2-u_1)}$$

$$0.5605 - v_f = \frac{(0.5605-0.6761)(508.5-500)}{(508.5-485.1)}$$

$$0.5605 + 0.04199 = \boxed{v_f = 0.6025 \ ft^3/lb}$$

80% Effort

(b) for Ammonia $T = 40°F$ and $x = 0.5$, determine Pressure in lbf/in^2, v in ft^3/lb and u in BTU/lb.

finding pressure at $T = 40°F$ in liquid vapor state

$T_2 = 41.11$, $T_1 = 37.67$, $T_x = 40$ $(°F)$ $\quad P_2 = 75$, $P_1 = 70$, $P_x =$ find $\left(lbf/in^2\right)$

$$\frac{T_2-T_1}{T_2-T_x} = \frac{P_2-P_1}{P_2-P_f}$$

$$P_2 - P_f = \frac{(P_2-P_1)(T_2-T_x)}{(T_2-T_1)}$$

$$75 - P_f = \frac{(75-70)(41.11-40)}{(41.11-37.67)}$$

$$75 - P_f = 1.6133$$

$$75 - 1.6133 = \boxed{P_f = 73.387 \ lbf/in^2}$$

at $T = 40°F$ finding Specific volume for gas and liquid state.

for liquid

$T_2 = 41.11$, $T_1 = 37.67$, $T_x = 40$ $(°F)$ $\quad V_2 = 0.02536$, $V_1 = 0.02526$, $V_x =$ find $\left(ft^3/lb\right)$

$$\frac{T_2-T_1}{T_2-T_x} = \frac{V_2-V_1}{V_2-V_x}$$

$$V_2 - V_x = \frac{(V_2-V_1)(T_2-T_x)}{(T_2-T_1)}$$

$$0.02536 - V_x = \frac{(0.02536-0.02526)(41.11-40)}{(41.11-37.67)}$$

$$0.02536 - 3.2267\times10^{-5} = \boxed{V_x = 0.02533 \ ft^3/lb}$$ liquid

for gas

$T_2 = 41.11$, $T_1 = 37.67$, $T_x = 40$ $(°F)$ $\quad V_2 = 3.8837$, $V_1 = 4.1473$, $V_x =$ find

$$V_2 - V_x = \frac{(V_2-V_1)(T_2-T_x)}{\ldots}$$

$$V_{mix} = V_{fluid} + x\left(V_{gas} - V_{fluid}\right)$$

$(\ldots 0.02533)$

$= 0.02533 + 0.5(3.8838 - 0.02533)$

$= 0.02533 + 1.9292$

$\boxed{V_{mix} = 1.9546\ ft^3/lb}$

$$V_2 - V_x = \frac{(V_2 - V_1)(T_2 - T_x)}{(T_2 - T_1)}$$

$$3.8837 - V_x = \frac{(3.8837 - 4.1473)(41.11 - 40)}{(41.11 - 37.67)}$$

$3.8837 + 7.85\times10^{-5} = \boxed{V_x = 3.8838\ ft^3/lb}$

Specific vol of gas state.

External Energy (U)

for liquid state at T 40°F, liquid vap state

$T_2 = 41.11$, $T_1 = 37.67$, $T_x = 40$ (°F)

$u_2 = 87.21$, $u_1 = 83.40$, $u_f =$ find (BTU/lb)

$$\frac{T_2 - T_1}{T_2 - T_x} = \frac{u_2 - u_1}{u_2 - u_x}$$

$$u_2 - u_f = \frac{(u_2 - u_1)(T_2 - T_x)}{(T_2 - T_1)}$$

$$87.21 - u_f = \frac{(87.21 - 83.40)(41.11 - 40)}{(41.11 - 37.67)}$$

$87.21 - 1.2294 = \boxed{u_f = 85.981\ BTU/lb}$

for gas

$u_2 = 568.73$, $u_1 = 567.99$ (BTU/lb), $u_f =$ find

$$u_2 - u_f = \frac{(u_2 - u_1)(T_2 - T_x)}{T_2 - T_1}$$

$$568.73 - u_f = \frac{(568.73 - 567.99)(41.11 - 40)}{(41.11 - 37.67)}$$

$568.73 - 0.2388 = \boxed{u_f = 568.49\ BTU/lb}$

Noh

(c) for water P = 80 bar and V = 0.04034 m³/kg, determine Temp at °C, u in kJ/kg and h is kJ/kg.

Sol: at P = 80 bar, state liquid vapor

$P_2 = 85.81$, $P_1 = 74.36$, $P_x = 80$ (bar)

$T_2 = 300$, $T_1 = 290$, $T_{final} = ?$

$$\frac{P_2 - P_1}{P_2 - P_x} = \frac{T_2 - T_1}{T_2 - T_f}$$

$$T_2 - T_f = \frac{(T_2 - T_1)(P_2 - P_x)}{(P_2 - P_1)}$$

$$300 - T_f = \frac{(300 - 290)(85.81 - 80)}{(85.81 - 74.36)}$$

$300 - 5.0742 = \boxed{T_f = 294.93\ °C}$

at P 80 bar External Energy,

for liquid

$P_2 = 85.81$, $P_1 = 74.36$, $P_x = 80$ (Bar)

$u_2 = 1332$, $u_1 = 1278.9$, $u_{final} =$ find (kJ/kg)

for gas

$P_2 = 85.81$, $P_1 = 74.36$, $P_x = 80$ (Bar)

$u_2 = 2563$, $u_1 = 2576$, $u_f =$ find (kJ/kg)

$P_2 - P_1$ ... $u_2 - u_1$

$$u_2 - u_f = \frac{(u_2 - u_1)(P_2 - P_x)}{}$$

$$\frac{P_2-P_1}{P_2-P_x} = \frac{u_2-u_1}{u_2-u_x}$$

$$u_2-u_f = \frac{(u_2-u_1)(P_2-P_x)}{(P_2-P_1)}$$

$$1332-u_f = \frac{(1332-1278.9)(85.81-80)}{(85.81-74.36)}$$

$$1332-26.9442 = \boxed{u_f = 1305.1 \text{ kJ/kg}}$$

$$u_2-u_f = \frac{\ldots}{(P_2-P_1)}$$

$$2563-u_f = \frac{(2563-2576)(85.81-80)}{(85.81-74.36)}$$

$$2563+6.5965 = \boxed{u_f = 2569.597 \text{ kJ/kg}}$$

for Specific volume of gas $v = 0.04034\ m^3/kg$, finding h for gas, liquid

gas

$v_2 = 0.03944$
$v_1 = 0.04406$ (m^3/kg)
$v_x = 0.04034$

$h_2 = 1640.1$
$h_1 = 1676.4$
h_{final} = find (kJ/kg)

$$\frac{v_2-v_1}{v_2-v_x} = \frac{h_2-h_1}{h_2-h_f}$$

$$h_2-h_f = \frac{(h_2-h_1)(v_2-v_x)}{(v_2-v_1)}$$

$$1640.1-h_f = \frac{(1640.1-1676.4)(0.03944-0.04034)}{(0.03944-0.04406)}$$

$$1640.1+7.0714 = \boxed{h_f = 1647.17 \text{ kJ/kg}}$$

liquid,

$h_2 = 1154.2$
$h_1 = 1121.9$ (kJ/kg)
h_f = find

$$h_2-h_f = \frac{(h_2-h_1)(v_2-v_x)}{(v_2-v_1)}$$

$$1154.2-h_f = \frac{(1154.2-1121.9)(0.03944-0.04034)}{(0.03944-0.04406)}$$

$$1154.2-6.29 = \boxed{h_f = 1147.91 \text{ kJ/kg}}$$

80

(d) for Refrigerant $T = -20°C$ and $h = 235.31$ kJ/kg, determine p in bar, v in m^3/kg and u in kJ/kg

at $-20°C$, Pressure is 1.3299 bar
and h = 235.31 kJ/kg Specific vol for liquid = 0.7361 m^3/kg
for gas = 0.1464 m^3/kg
Internal Energy (u) for liquid = 24.17 kJ/kg
for gas = 215.84 kJ/kg

Question #2 Water Contain in a Closed, rigid tank, initially

Saturated vapor at 180°C, is cooled to 100°C

(a) Determine the initial pressure, in MPa.

(b) Determine the final quality.

Sol^n Rigid tank = volume stay constant

saturated vapor state → Specific volume V_{mix}

$v_1 = v_2$

$V_x = 0.1941\ m^3/kg$

$\boxed{\text{Temp } 180°C}$ → Pressure ?

$T_2 = 198.3$ (°C)
$T_1 = 179.9$
$T_x = 180$

$P_2 = 15$
$P_1 = 10$
$P_f = \text{find}$

$$\frac{T_2 - T_1}{T_2 - T_x} = \frac{P_2 - P_1}{P_2 - P_x}$$

$$P_2 - P_x = \frac{(P_2 - P_1)(T_2 - T_x)}{T_2 - T_1}$$

$$15 - P_x = \frac{(15-10)(198.3 - 180)}{(198.3 - 179.9)}$$

$15 - 4.973 = P_x = 10.027$ bar

1 bar = 0.1 Mpascal

10.027 bar = (0.1 × 10.027) MPa

P = 1.0027 MPa

(b) final quality

$$V_{mix} = V_{fluid} - X(v_{fluid} - V_{gas})$$

at 100°C

$T_2 = 111.4$ (°C)
$T_1 = 99.63$
$T_x = 100$

v liquid
$U_2 = 1.0528$
$V_1 = 1.0432$
$U_f = \text{find}$

v gas
$V_2 = 1.159$
$V_1 = 1.694$
$V_f = \text{find}$

Liquid

$$\frac{T_2 - T_1}{T_2 - T_x} = \frac{V_2 - V_1}{V_2 - V_f}$$ vol m³/kg

$$V_2 - V_f = \frac{(V_2 - V_1)(T_2 - T_x)}{(T_2 - T_1)}$$

$$1.0528 - V_f = \frac{(1.0528 - 1.0432)(111.4 - 100)}{(111.4 - 99.63)}$$

$1.0528 - 0.009298 = V_f = 1.0435\ m^3/kg$

for gas

$$V_2 - V_f = \frac{(V_2 - V_1)(T_2 - T_x)}{(T_2 - T_1)}$$

$$1.159 - V_f = \frac{(1.159 - 1.694)(111.4 - 100)}{(111.4 - 99.63)}$$

$1.159 + 0.5182 = V_f = 1.6772\ m^3/kg$

$V_x = V_{mix} = 0.1941\ m^3/kg$

$$V_{mix} = V_{fluid} - X(V_{fluid} - V_{gas})$$

$(0.1941\ m^3$ ... $1.0435 - x(1.0435 - 1.6772)$

$(0.1941)\ m^3/kg : 1.0435$ ...

$$\frac{0.1941 - 1.0435}{(1.0435 - 1.6972)} = \boxed{x = 1.34}$$

80%

Question-3

2 kg of water vapor, constant pressure = 300 kPa (3 bar), from a saturated vapor state, volume $2.064\ m^3$

(a) initial temperature in °C at pressure 3 bar

State saturated vapor

$P_1 = 3.613$ (bar) $\quad T_1 = 140$
$P_2 = 2.701 \quad T_2 = 130$
$P_x = 3 \quad T_x = \text{find}$

P_1, P_2, T_1, T_2 from Table

$$\frac{P_1 - P_2}{P_1 - P_x} = \frac{T_1 - T_2}{T_1 - T_x}$$

$$T_1 - T_x = \frac{(T_1 - T_2)(P_1 - P_x)}{(P_1 - P_2)}$$

$$140 - T_x = \frac{(140 - 130)(3.613 - 3)}{(3.613 - 2.701)}$$

$$140 - 6.72 = \boxed{T_x = 133.28°C} \approx 134°$$

(b) Determine final Temperature.

State 2, Superheated

at the constant Pressure, but volume has changed. $V = 2.064\ m^3$

$$\text{Specific vol} = \frac{V}{m} = \frac{2.0643\ m^3}{2\ kg} = 1.03215\ m^3/kg$$

Pressure is 3 bar

$\boxed{\text{Temperature is } 400°C}$

(c) Determine the work in the process

$$\text{work} = -\int p\,dv \quad \text{at pressure 3 bar}$$

$= -(3\times10^5 (kPa))(1.03215 - 1.2332)$

$= 60315\ J$ or $60.315\ kJ$

at pressure 3 bar in state 1 and volume = ?

Initial volume: find

$P = 3$ bar
$T = 134$

$T_2 = 160 \quad v_2 = 1.317$
$T_1 = 120 \quad v_1 = 1.188$
$T_x = 134^\circ \quad v_x = \text{find}$

$$\frac{T_2 - T_1}{T_2 - T_x} = \frac{v_2 - v_1}{v_2 - v_x}$$

$$v_2 - v_x = \frac{(v_2 - v_1)(T_2 - T_x)}{(T_2 - T_1)}$$

$$1.317 - v_x = \frac{(1.317 - 1.188)(160 - 134)}{(160 - 120)}$$

$1.317 - 0.08385 = \boxed{v_x = 1.2332}$

4 Refrigerant 134 a

$P_1 = 10$ bar $\quad T_1 = 60^\circ C$
$P_2 = 2$ bar $\quad T_2 = 30^\circ C$

$M_{refrigerant} = 1$ kg

work done by the refrigerant is 28 kJ

Potential and kinetic energy effects are negligible

(a) Determine the initial vol of refrigerant, V_1, m^3.

from 134 a index

at, $P = 10$ bar and $T = 60^\circ C$

Specific volume $0.02301\ m^3/kg$,

Initial volume $= v \times m = 0.02301\ m^3/kg \times 1\ kg$

$V_1 \Rightarrow \boxed{= 0.02301\ m^3}$

at this time internal energy $U_1 = 268.35\ kJ/kg$

and $T = 30^\circ C$

at Pressure 2bar and ...

Specific volume $= 0.11856\ m^3/kg$

Occupied vol $= v \times m = 0.11856\ m^3/kg \times 1\,kg$

$= 0.11856\ m^3$

Internal energy at this time

$u_2 = 253.06\ kJ/kg$

(b) Determine the heat transfer, while work done is given by 20 kJ

$\Delta U = Q - W$

$(U_2 - U_1) + W = Q$

$= (253.06 - 268.35)\,kJ + 20\,kJ = Q$

$-15.29\,kJ + 20\,kJ$ $Q = 4.71\,kJ$

$U_2 = u_2 \times m = 253.06\ \frac{kJ}{kg} \times 1\,kg$

$= 253.06\,kJ$

$U_1 = u_2 \times m = 268.35\ \frac{kJ}{kg} \times 1\,kg$

$= 268.35\,kJ$

(IF5) A closed, Rigid tank 1.5 kg of water, initially two phases liq-vap mixture at $T_1 = 70^\circ C$.
Heat transfer occurs until the tank contains only saturated vapor at $T_2 = 120^\circ C$

(a) Determine initial Pressure.

State 1, Saturated liq-vapor

at 70°C, pressure is 0.3119 bar or $(0.3119 \times 10^5$ Pascal$) = 31.19\ kPa$

$v_g = 5.042\ m^3/kg$

$\boxed{u_1 = 2469.6\ \frac{kJ}{kg}}$

(b) State 2 Saturated vapor at $t = 120^\circ C$

at $t = 120^\circ C$, $P = 1.985$ bar, $u_2 = 2529.3\ \frac{kJ}{kg}$

rigid tank: $v_1 = v_2 = 5.042\ m^3/kg$

$\Delta U = (u_2 - u_1)\,m = (2529.3 - 2469.6)\ \frac{kJ}{kg} \times 1.5\,kg$

$= 89.55\,kJ$

$Q = \Delta U$
$Q = 89.55 \text{ kJ}$

\#6 Tank contain two phase liquid-vapor mixture of refrigerant 22 at 10 bar. The mass of saturated liquid in the tank is 25 kg and the quality is 60%.

(a) Determine the volume of the tank in m^3.

Sol^n State 1, liq-vapor mixture

at $P = 10$ bar, Temperature is $23.40°C$,

$v_f = 0.8352 \times 10^{-3}$
$v_g = 0.0236$

$x = 60$ 60%

$v_{mix} = v_f + x(v_g - v_f)$

$(0.08352 \times 10^{-3}) + 60(0.0236 - 0.8352 \times 10^{-3})$

$\boxed{v_{mix} = 1.36597 \text{ m}^3/\text{kg}}$

occupied $V = v \times m$
$= (1.36597) \frac{m^3}{kg} \times 25 \text{ kg}$
$\boxed{= 34.15 \text{ m}^3}$

80%

(b) Determine the fraction of the total volume occupied by saturated vapor.

mass of liquid (25 kg) and quality 60%.

$\boxed{mass_{(l)} = x \cdot (m) = 25(60) \text{ kg} \quad 1500 \text{ (kg)}}$ (60%)

$mass_{(g)} = x(m) = 25(40) = 1000 \text{ kg}$ (40%)

Quality can't be $> \boxed{100\%}$

Specific vol

$v_1 = \frac{V}{m} = \frac{34.15 \text{ m}^3}{1500 \text{ kg}} = 0.02277 \text{ m}^3/\text{kg}$

$v_2 = \frac{V}{m} = \frac{34.15 \text{ m}^3}{1000 \text{ kg}} = 0.03415 \text{ m}^3/\text{kg}$

$v_{mix} = v_f + x(v_g - v_f)$

$\frac{v_{mix} - v_f}{v_g - v_f} = x = \frac{(1.36597 - 0.02277) \text{ m}^3/\text{kg}}{(0.03415 - 0.02277) \text{ m}^3/\text{kg}} = 118$

\#7 A balloon filled with 0.2 grams of helium, initially at 27°C, 1.3 bar, is released and rises in the atmosphere until the helium is at 17°C, 0.80 bar. Assume that the helium behaves as an ideal gas.

behaves ...

(a) Determine R, in $\left(\frac{J}{g \cdot K}\right)$

$R = \frac{\bar{R}}{m}$ $\quad \bar{R} = 8.314 \frac{kJ}{k.m \cdot {}^\circ K}$

$m = 4.003 \frac{kg}{kmol}$

$$R = \frac{8.314 \frac{J}{mol \cdot K}}{4.003 \frac{kg}{kmol}} = 2.0769 \left[\frac{J}{g \cdot K}\right]$$

$\frac{kg \cdot m}{s^2}$

(b) Determine initial volume, in cm^3?

initially, $T = 27^\circ$ $P = 1.3$ bar

$$V_1 = \frac{mRT}{P} = \frac{\left(\frac{0.2}{4.003}\right)\cancel{mol}\ 8.314 \frac{J}{\cancel{mol} \cdot \cancel{K}} \times (27+273)\cancel{K}}{(1.3 \times 10^5 \text{ Pascal})\ \frac{kg}{m \cdot s^2}}$$

$= 9.586 \times 10^{-4}\ m^3$ to cm^3

$\rightarrow 1m^3 = 1 \times 10^6\ cm^3$

$9.522 \times 10^{-4} m^3 = 1 \times 10^6 \times 9.522 \times 10^{-6}\ cm^3$

$\boxed{\approx 959\ cm^3}$

(b) final volume $P = 0.80$ bar, $T = 17^\circ C$

$$V_f = \frac{mRT}{P} = \frac{\left(\frac{0.2}{4.003}\right)(8.314)(17+273)}{(0.80 \times 10^5)} = m^3$$

$= 0.0015057 8\ m^3$ to cm^3 $\times 10^6$

$\approx 1506\ cm^3$

(8) CO_2, fixed rigid tank, (volume constant)

initially $25^\circ C$, 2.75 bar, volume $0.05 m^3$.

gas stirred until its temp is $257^\circ C$, heat transfer, surrounding occurs, 2.75 KJ.

(a) Determine the mass of the CO_2, in kg.

$$R = \frac{\bar{R}}{\text{molar mass of } CO_2} = \frac{8.314}{44.01} = 0.1889\ KJ/kmol\text{-}K$$

mass of CO_2 = $\frac{PV}{RT}$ (Ideal gas law)

$$= \frac{2.75 \times 10^5 (Pa) \times (0.05) m^3}{(0.1889) J/mol \times (25+273) K} = \boxed{0.2398\ kg}$$

mass of CO_2

44.01 kg/kmol

at temp 298 kelvin $\bar{U}_1 = 6685\ \frac{KJ}{kmol}$

at temp $(257°C + 273)$ = 530K $\bar{U}_2 = 14622\ \frac{KJ}{kmol}$

(b) Determine the work, in KJ.

$W + Q = \Delta U$

$W = \Delta U - Q$

$= 180.345 - 2.75$

$\boxed{\text{work} = 177.60\ KJ}$

$\Delta U = U_2 - U_1 = 14622 - 6685$

$= 7937$ kJ/kmol

$\Delta U = \frac{7937}{44.01} = 180.345$ KJ

80%